CENG381 Stochastic Processes

Rate of Convergence - Answer Key 1

Q1 Solution

a) Stationary distribution ($p=0.3$, $q=0.4$):

$\pi_E = q/(p+q) = 0.4/0.7 = 4/7 \sim 0.571$

$\pi_W = p/(p+q) = 0.3/0.7 = 3/7 \sim 0.429$

b) Convergence derivation:

$\Delta_t = \mu_t(E) - \pi_E$

$$\begin{aligned}
\mu_{t+1}(E) &= \mu_t(E)(1-p) + \mu_t(W) q \\
&= \mu_t(E)(1-p) + (1-\mu_t(E)) q \\
&= \mu_t(E)(1-p-q) + q
\end{aligned}$$

So
$$\begin{aligned}
\Delta_{t+1} &= \mu_{t+1}(E) - \pi_E \\
&= \mu_t(E)(1-p-q) + q - q/(p+q) \\
&= (1-p-q)\mu_t(E) + q(1 - 1/(p+q)) \quad \text{-- rearranged:} \\
&= (1-p-q)(\mu_t(E) - q/(p+q)) \\
&= (1-p-q)\Delta_t
\end{aligned}$$

Rate of convergence: $\lambda_2 = 1 - p - q = 1 - 0.3 - 0.4 = 0.3$

c) Number of steps needed:

$\Delta_0 = \mu_0(E) - \pi_E = 1 - 4/7 = 3/7 \sim 0.4286$

$\Delta_t = (0.3)^t \cdot (3/7)$

We need $(0.3)^t \cdot (3/7) < 0.01$

$(0.3)^t < 0.01 \cdot 7/3 = 0.0233$

$t \cdot \ln(0.3) < \ln(0.0233)$

$t > \ln(0.0233)/\ln(0.3) = (-3.757)/(-1.204) \sim 3.12$

So $t \geq 4$ steps are sufficient. Verify:

$t=4$: $(0.3)^4 \cdot 3/7 = 0.0081 \cdot 0.4286 \sim 0.0035 < 0.01$ (OK)

Q2 Solution

a) For $p=0.2$, $q=0.5$:

$\lambda_1 = 1$, $\lambda_2 = 1 - 0.2 - 0.5 = 0.3$

D = diag(1, 0.3)

Eigenvector for lambda_1=1: stationary dist row = [q/(p+q), p/(p+q)]
 = [0.5/0.7, 0.2/0.7] = [5/7, 2/7]
Eigenvector for lambda_2=0.3: [p, -q] direction (up to scaling) = [1, -1]

$$S = \begin{bmatrix} q/(p+q) & p/(p+q) \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5/7 & 2/7 \\ 1 & -1 \end{bmatrix}$$

<u>b) Explicit P^n formula:</u>

P^n = S^(-1) * D^n * S

This gives:
 p_{EE}^(n) = q/(p+q) + lambda_2^n * p/(p+q)
 p_{EW}^(n) = p/(p+q) - lambda_2^n * p/(p+q)
 p_{WE}^(n) = q/(p+q) - lambda_2^n * q/(p+q)
 p_{WW}^(n) = p/(p+q) + lambda_2^n * q/(p+q)

As n->inf, lambda_2^n -> 0, so all rows converge to [q/(p+q), p/(p+q)].

<u>c) P^(33)_{EE} for p=0.2, q=0.5:</u>

p_{EE}^(33) = q/(p+q) + lambda_2^33 * p/(p+q)
 = 5/7 + (0.3)^33 * (2/7)

(0.3)^33 is extremely small (~ 5.6 x 10^(-18)), so:
p_{EE}^(33) ~ 5/7 ~ 0.7143 (the chain has essentially converged)

Q3 Solution

<u>a) Lazy chain Q = (1/2)I + (1/2)P:</u>

$$\begin{array}{cc} & \begin{array}{cc} E & W \end{array} \\ \begin{array}{c} E \\ W \end{array} & \begin{bmatrix} 0.80 & 0.20 \\ 0.15 & 0.85 \end{bmatrix} \end{array}$$

Check: 0.5*1 + 0.5*0.6 = 0.8, 0.5*0 + 0.5*0.4 = 0.2 (OK)

<u>b) Eigenvalues of Q:</u>

lambda_1(Q) = 1 (every stochastic matrix has eigenvalue 1) lambda_2(Q) = (1/2)*1 + (1/2)*lambda_2(P)

= 1/2 + (1/2)*(1 - 0.4 - 0.3)

= 1/2 + (1/2)*(0.3)

= 1/2 + 0.15 = 0.65

Original lambda_2(P) = 1 - 0.4 - 0.3 = 0.3

Lazy chain lambda_2(Q) = 0.65 > 0.3

c) Comparison and motivation:

The lazy chain (lambda_2=0.65) converges SLOWER than the original (lambda_2=0.30), because the step size is halved each iteration.

However, the key advantage is APERIODICITY. A chain with period d>1 does not converge to pi (it oscillates). Adding self-loops (laziness) breaks any periodic structure, guaranteeing aperiodicity and hence true convergence to pi. The tradeoff is slower (but guaranteed) convergence vs. possible non-convergence without laziness.